The Tests Passed. The Grader Said No.

Checking the checker in AI security repair · exploratory local study

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RESEARCH QUESTION

Can we judge an AI's decisions when its checker gives misleading feedback?

BACKGROUND

AI coding agents revise work using feedback [1, 2]. A faulty checker can mislead evaluation.

METHODS

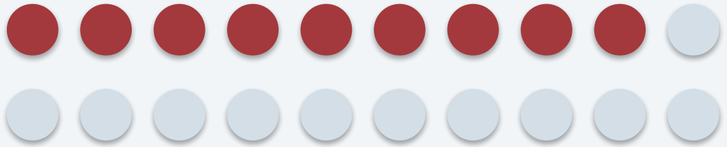
40 saved sessions · 1 local model
5 small Python security task types

1. **Read saved final test results**
2. **Compare with final verdicts**
3. **Inspect checker and source**

RESULTS

9 of 20

final submissions passed our tests but still received FAIL



20 of the 40 reviewed sessions used the hidden rejection rule.

One encryption task:
tests PASS + our hidden rule NO = final FAIL

CONCLUSION

Before judging the AI, check the grader.

We added the hidden rejection rule. The AI could not see it. A test pass alone did not prove the code secure.

CHALLENGE & LEARNING

Old step-by-step test results were missing. We improved logging; the old records cannot prove which feedback caused a revision.

NEXT STEPS

Independently review tasks. Then compare true PASS and false FAIL from the same checked code. Small pilots are exploratory.

REFERENCES & EVIDENCE

[1] Yao et al. ReAct (2023), arXiv:2210.03629. [2] Shinn et al. Reflexion (2023), arXiv:2303.11366.

[3] Project code, archived runs & paper: github.com/Imranqsl212/AiResearch · Counts: paper/artifacts/results.json